

PHYS 427: Lecture # 1

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Lecture Span

1. Probability and Multiplicity

Notes

1. Homework due Mondays at 11am in the homework box in interpass (427)
2. Discussion is on Tuesdays, weekly quizzes
3. 2 Midterms (October, November) in class, One final

Microstates and Macrostates

Definition 0.1. A Microstate is the complete description of every particle or configuration of a system, down to every detail.

Definition 0.2. A Macrostate is a description of a few measurable properties of a configuration.

In this class, we will review gas, solids, liquids, and magnets.

Example 1: Consider a few gas molecules in a box, then each particle has a position and a momentum. Then, a microstate will be a description of all the atoms position and momentum, i.e.

$$\text{Microstate} = \{x_i, p_i\} \quad (1)$$

The macrostate will be

$$\text{Macrostate} = p, T, V \quad (2)$$

Or pressure, temperature, and volume.

Example 2: Consider a polymer that is a long chain of several other molecules. Then

$$\text{Microstate} = \{\text{configuration of each polymer}\} \quad (3)$$

$$\text{Macrostate} = \{\text{Distance from end to end of polymer chain}\} \quad (4)$$

Example 3: Consider a magnet with spins up and down in various configurations. This is an Ising Model. Then, the microstate would be

Microstate = Configuration of every spin in the magnet

Macrostate = Average magnetization (average of all the spin values), M

Example

The most probable macrostate is the one who has the most degenerate microstates. Consider a box divided into a left half and a right half. Consider the fact that we have N particles in this box. Let the Macrostate be the number of particles in the left half n_L . Then $n_R = N - n_L$.

Microstate: Configuration of N particles as a list of L/R state of each particle. That is, whether a particle is in the left or right half. Like

LRRLRRL...

Consider multiplicity $\Omega(n_L, N)$. Then consider a table with the Microstate on the left column, Ω in the middle, and Macrostate on the right.

Here, $N = 1$

Microstate	Ω	Macrostate
Particle on the right	1	$n_L = 0$
Particle on the left	1	$n_L = 1$

Total number of microstates is $2 = 2^1$, and total number of macrostates is 1.

For $N = 2$

Microstate	Ω	Macrostate
2 particles on the right	1	$n_L = 0$
Symmetric particle distribution	2	$n_L = 1$
2 particles on the left	1	$n_L = 2$

Total number of microstates is $4 = 2^2$, and macrostates is $3 = 2 + 1$.

For $N = 3$,

Microstate	Ω	Macrostate
3 particles on the right	1	$n_L = 0$
1 particle on the left	3	$n_L = 1$
1 particle on the right	3	$n_L = 2$
3 particles on the left	1	$n_L = 3$

Total number of microstates is $8 = 2^3$, and macrostates is $4 = 3 + 1$.

Definition 0.3. A *Statistical Ensemble* is the total collection of all possible microstates of a system.

The total number of microstates is 2^N and the total number of macrostates is $N + 1$. This is for a system with only two states.

Note, for small N , the probability of measuring a specific macrostate will not be that much different from each other. All the tools we develop in this course work in the Thermodynamic limit, that is $N \rightarrow \infty$.

Next, we derive the general formula for the multiplicity. For the first particle, we have N choices. For the 2nd particle, we have $N - 1$ choices. Etc. In general, for the n -th particle, we have $N - n + 1$ choices.

Then, the number of choice is $N(N - 1)(N - 2) \dots (N - n_L + 1)$ if there are only L slots.

Here, the general formula for the multiplicity is

$$\Omega(n_L, N) = \frac{N(N - 1) \dots (N - n_L + 1)}{n_L!} = \frac{N!}{n_L!(N - n_L)!} = \binom{N}{n_L} \quad (5)$$

This is also N "choose" n_L .

The reason why we divide the number of choices by $n_L!$ is because, since all the particles are identical, we can rearrange them in $n_L!$ possible configurations in the left box, but we would get back the same configuration since all the particles are identical.

Moreover, using the binomial theorem which states

$$(p + q)^N = \sum_{n=0}^N \binom{N}{n} p^n q^{N-n} \quad (6)$$

$$2^N = \sum \binom{N}{n_L} = \sum \Omega(n_L, N) \quad (7)$$

Fundamental Assumption: In a closed, isolated system (i.e., non-interacting with surroundings), each microstate in a system is equally likely. Therefore, the probability of measuring one particle microstate is $1/2^N$. But some macrostates are more likely than others.

Therefore, the probability that n_L out of N particles are to the left is

$$P(n_L, N) = \frac{\Omega(n_L, N)}{\sum_{n_L} \Omega(n_L, N)} \iff \text{total \# of states} \quad (8)$$

As $N \rightarrow \infty$, $P(n_L, N) \rightarrow \delta_{N/2}$, which is the delta function located at $N/2$ particles on the left and right. Check that

$$\sum P = 1 \quad (9)$$

$$\langle n_L \rangle = \frac{N}{2} \quad (10)$$

$$\langle f(n_L) \rangle = \sum f(n_L) P(n_L) \quad (11)$$